Cryptography – Information Security

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SCC.160:
Fundamentals of Communication Systems

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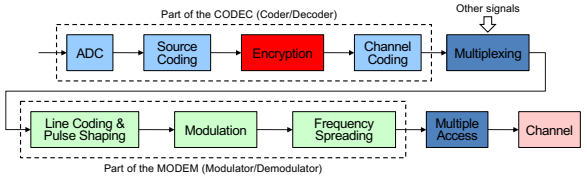
In Previous Lecture, we talked about Block diagram of a transmitter



- **Analogue-to-Digital Converter (ADC)**: Sampling, quantisation and pulse-code modulation (PCM).
- **Source coding**: Removal of redundancy (compression).
- **Encryption**: Employment of cipher to increase information security.
- **Channel Coding**: Introduction of redundant bits for error control (detection and correction).

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Introduction

- There are three security goals:
- Confidentiality, keeping information secret from unauthorized access, is probably the most common aspect of information security
- Integrity, Information needs to be changed constantly. Integrity means that changes should be done only by authorized users and through authorized mechanisms. Protected from unauthorized change

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Cont'd

- Availability, The information created and stored by an organization needs to be available to authorized users and applications. Information is useless if it is not available.

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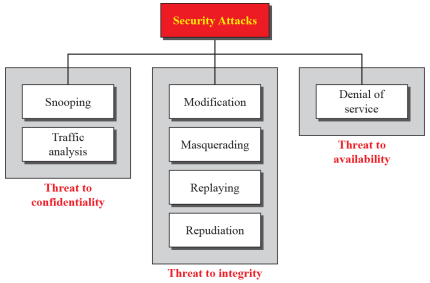
Attacks

- The three goals of security—*confidentiality*, *integrity*, and *availability*—can be threatened by following security attacks.

Snooping 窥探

Masquerading 伪装

Repudiation 否认



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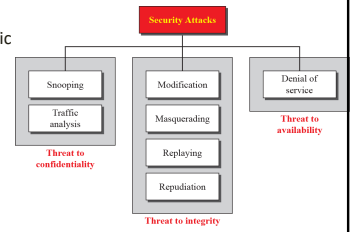
Attacks (cont'd)

- Attacks threatening confidentiality
  Two types of attack threaten the confidentiality of information: snooping 窥探 and traffic analysis. Snooping refers to unauthorized access to or interception of data. Traffic analysis refers other types of information collected by an intruder by monitoring online traffic.
- Attacks threatening integrity
  The integrity of data can be threatened by several kinds of attack: *modification*, *masquerading*, *replaying*, and *repudiation*.
- Attacks threatening availability
- We mention only one attack threatening availability: *denial of service*.



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Attacks Threatening Integrity

Modification means that the attacker intercepts the message and changes it.

Masquerading or spoofing happens when the attacker impersonates somebody else.

Replaying means the attacker obtains a copy of a message sent by a user and later tries to replay it.

Repudiation means that sender of the message might later deny that she has sent the message; the receiver of the message might later deny that he has received the message.

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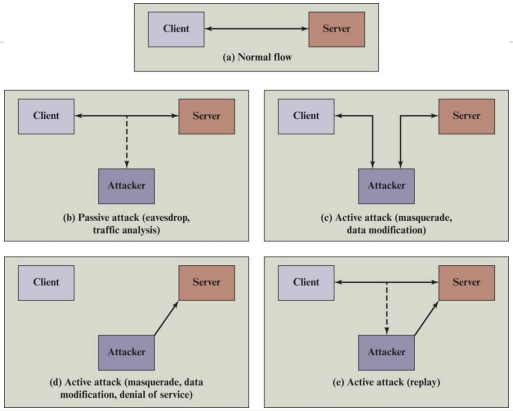
Security Attacks



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Services and Techniques

- Two techniques are prevalent today: one is very general (cryptography) and one is specific (steganography).

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Message Security - Encryption

- Two techniques are dominant today : cryptography and steganography.
- Cryptography, a word with Greek origins, means "secret writing."
  - The science of transforming messages to make them secure and immune to attacks
- The word steganography, with origin in Greek, means "covered writing," in contrast with cryptography, which means "secret writing."

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Steganography

- The strategy for secret communication relied on simply **hiding** the message **from sight**.
- Other examples include:
  - Shave head, write message on scalp, re-grow hair (Greece)
  - Write message on silk, roll it into a ball, wax it, swallow it (China)
  - Make special ink and write on the shell of a boiled egg (Italy)
  - Use invisible ink that turns brown when heated (Italy)
- Secret communication which is achieved by hiding the existence of a message is known as **steganography**.
- **Fundamental weakness**: Interception of the message compromises all security.

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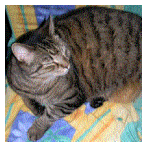
Modern Steganography

Color (Green)	Base 10	Binary
	238	11101110
	235	11101011
	232	11101000



The hidden image is revealed by removing all but the two least significant bits of each colour component and a subsequent normalization.

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Cover Image

Image to Merge

Merged Image

Unmerged Image

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Cryptography

- The aim of cryptography is not to hide the existence of a message but rather to **hide its meaning** - a process known as **encryption**.
- A protocol, which is agreed between the sender and the receiver, is used to scramble messages and render them unintelligible.
- Cryptography can be divided into transposition and substitution.
- In **transposition**, the letters of a message are simply rearranged.
- In **substitution**, individual characters or blocks of characters (e.g., words) are substituted by different characters or symbols.

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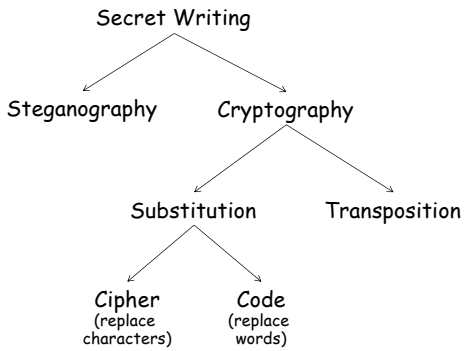
Branches of 'secret writing'



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Definitions (1 of 2)

- Plaintext
  - An original message
- Ciphertext
  - The coded message
- Enciphering/encryption
  - The process of converting from plaintext to ciphertext
- Deciphering/decryption
  - Restoring the plaintext from the ciphertext

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Definitions (2 of 2)

- Cryptography
  - The area of study of the many schemes used for encryption
- Cryptographic system/cipher
  - A scheme
- Cryptanalysis
  - Techniques used for deciphering a message without any knowledge of the enciphering details
- Cryptology
  - The areas of cryptography and cryptanalysis

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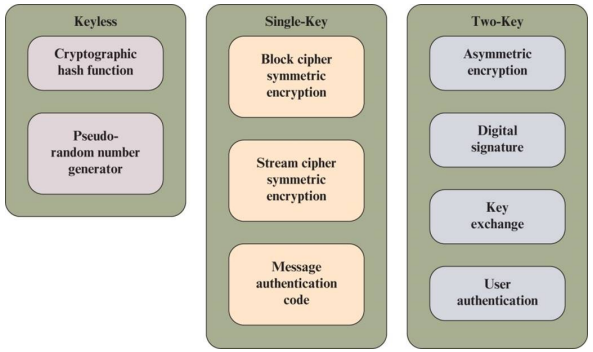
Cryptographic Algorithms



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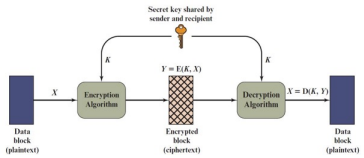
Cryptography

- Symmetric-Key Encipherment (加密)
  - Uses a single key for both encryption and decryption



- Asymmetric-Key Encipherment
  - Uses two keys instead of one.

Encryption: $C = E_k(P)$ Decryption: $P = D_k(C)$

In which, $D_k(E_k(x)) = E_k(D_k(x)) = x$

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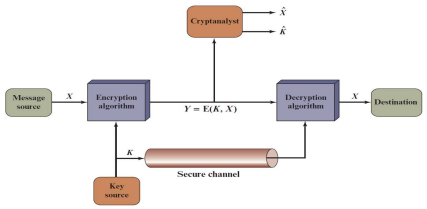
Symmetric Cipher Model

- There are two requirements for secure use of conventional encryption:
  - A strong encryption algorithm
  - Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure



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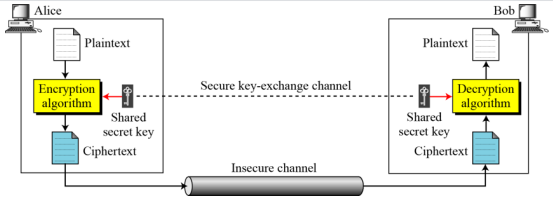
General idea of symmetric-key cipher



A symmetric-key cipher uses the same key for both encryption and decryption, and the key is used for bidirectional communication, that is why it is called symmetric.

To create the ciphertext from the plaintext, Alice uses an encryption algorithm and a shared secret key. To create the plaintext from ciphertext, Bob uses a decryption algorithm and the same secret key.

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Symmetric Key encryption

- Encryption can be thought of as locking the message in a box; decryption can be thought of as unlocking the box. In symmetric-key encipherment, the same key locks and unlocks, as shown in Figure below.

Locking and unlocking with the same key

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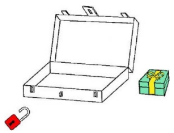
Activity

- Consider the case of asymmetric-key cryptography, where the sender owns a (red) padlock and a key, whilst the receiver owns a different (blue) padlock and a key. How should the sender have the briefcase delivered to the receiver without the requirement of key exchange?

Sender



Receiver

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Traditional symmetric-key ciphers

Traditional ciphers belong to the past. However, we briefly discuss them here because they can be thought of as the components of the modern ciphers. To be more exact, we can divide traditional ciphers into *substitution ciphers* and *transposition ciphers*.

A substiturion cipher replaces one symbol with another

The simplest mono-alphabetic cipher is the additive/shift/**Caeser** cipher.

The term additive cipher better reveals its mathematical nature.

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

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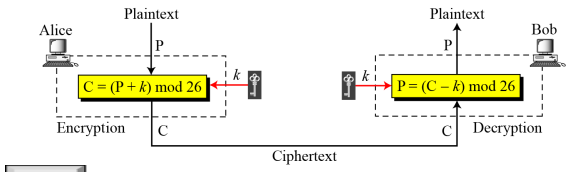

Note

When the cipher is additive, the plaintext, ciphertext, and key are integers in Z_{26}.

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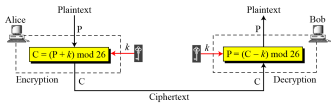
Example



Use the additive cipher with key = 15 to encrypt the message "hello".

Solution

We apply the encryption algorithm to the plaintext, character by character:

Plaintext: h → 07	Encryption: (07 + 15) mod 26	Ciphertext: 22 → W
Plaintext: e → 04	Encryption: (04 + 15) mod 26	Ciphertext: 19 → T
Plaintext: l → 11	Encryption: (11 + 15) mod 26	Ciphertext: 00 → A
Plaintext: l → 11	Encryption: (11 + 15) mod 26	Ciphertext: 00 → A
Plaintext: o → 14	Encryption: (14 + 15) mod 26	Ciphertext: 03 → D

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

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Decryption

Use the additive cipher with key = 15 to decrypt the message "WTAAD".

Solution

We apply the decryption algorithm to the plaintext character by character:

Ciphertext: W → 22	Decryption: (22 – 15) mod 26	Plaintext: 07 → h
Ciphertext: T → 19	Decryption: (19 – 15) mod 26	Plaintext: 04 → e
Ciphertext: A → 00	Decryption: (00 – 15) mod 26	Plaintext: 11 → l
Ciphertext: A → 00	Decryption: (00 – 15) mod 26	Plaintext: 11 → l
Ciphertext: D → 03	Decryption: (03 – 15) mod 26	Plaintext: 14 → o

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Brute-Force Cryptanalysis of Caesar Cipher

Eve has intercepted the ciphertext "UVACLYFZLJBYL". Show how she can use a brute-force attack to break the cipher.

Solution

Eve tries keys from 1 to 7. With a key of 7, the plaintext is "not very secure", which makes sense.

Ciphertext: UVACLYFZLJBYL

K = 1 → **Plaintext:** tuzbkxeykiaxk
K = 2 → **Plaintext:** styajwdxjhzwj
K = 3 → **Plaintext:** rsxzivcwigyvi
K = 4 → **Plaintext:** qrwyhubvhfxuh
K = 5 → **Plaintext:** pqvxgtaugewtg
K = 6 → **Plaintext:** opuwfsztfdvsf
K = 7 → **Plaintext:** notverysecure

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Frequency of characters in English

Letter	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*
E	12.7	H	6.1	W	2.3	K	0.08
T	9.1	R	6.0	F	2.2	J	0.02
A	8.2	D	4.3	G	2.0	Q	0.01
O	7.5	L	4.0	Y	2.0	X	0.01
I	7.0	C	2.8	P	1.9	Z	0.01
N	6.7	U	2.8	B	1.5		
S	6.3	M	2.4	V	1.0		

Frequency of diagrams and trigrams

Digram	TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF
Trigram	THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH

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Eve has intercepted the following ciphertext. Using a statistical attack, find the plaintext.

XLILSYWIMWRSAJSVWEPIJSVJSYVQMPPMSRHSPPEVWMXMWASVX-LQSVILY-VVCFIJSVIXLIWIPPIVVIGIMZIWQSVISJJIVW

Solution

When Eve tabulates the frequency of letters in this ciphertext, she gets: I =14, V =13, S =12, and so on. The most common character is I with 14 occurrences. This means key = 4.

the house is now for sale for four million dollars it is worth more hurry before the seller receives more offers

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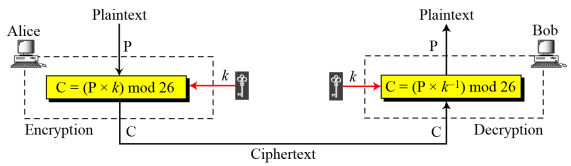
Multiplicative Ciphers



Note

In a multiplicative cipher, the plaintext and ciphertext are integers in Z_{26}; the key is an integer in Z26*.

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Addition and multiplication tables for Z_{10}

	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2
4	4	5	6	7	8	9	0	1	2	3
5	5	6	7	8	9	0	1	2	3	4
6	6	7	8	9	0	1	2	3	4	5
7	7	8	9	0	1	2	3	4	5	6
8	8	9	0	1	2	3	4	5	6	7
9	9	0	1	2	3	4	5	6	7	8

Addition Table in Z_{10}

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	0	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

Multiplication Table in Z_{10}

$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$Z_{10}^{*} = \{1, 3, 7, 9\}$

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Example

What is the key domain for any multiplicative cipher?

Solution

The key needs to be in Z_{26}*. This set has only 12 members: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25.

Example

We use a multiplicative cipher to encrypt the message "hello" with a key of 7. The ciphertext is "XCZZU".

Plaintext: h → 07	Encryption: (07 × 07) mod 26	ciphertext: 23 → X
Plaintext: e → 04	Encryption: (04 × 07) mod 26	ciphertext: 02 → C
Plaintext: l → 11	Encryption: (11 × 07) mod 26	ciphertext: 25 → Z
Plaintext: l → 11	Encryption: (11 × 07) mod 26	ciphertext: 25 → Z
Plaintext: o → 14	Encryption: (14 × 07) mod 26	ciphertext: 20 → U

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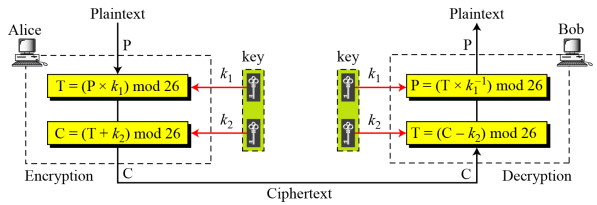
Affine cipher (additive + multiplicative)



$C = (P \times k_1 + k_2) \bmod 26$ $\qquad$ $P = ((C - k_2) \times k_1^{-1}) \bmod 26$

where k_1^{-1} is the multiplicative inverse of k_1 and $-k_2$ is the additive inverse of k_2

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The affine cipher uses a pair of keys in which the first key is from Z_{26}* and the second is from Z_{26}. The size of the key domain is 26 × 12 = 312.

Example

Use an affine cipher to encrypt the message "hello" with the key pair (7, 2).

P: h → 07	Encryption: (07 × 7 + 2) mod 26	C: 25 → Z
P: e → 04	Encryption: (04 × 7 + 2) mod 26	C: 04 → E
P: l → 11	Encryption: (11 × 7 + 2) mod 26	C: 01 → B
P: l → 11	Encryption: (11 × 7 + 2) mod 26	C: 01 → B
P: o → 14	Encryption: (14 × 7 + 2) mod 26	C: 22 → W

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Example

Use the affine cipher to decrypt the message "ZEBBW" with the key pair (7, 2) in modulus 26.

Solution

C: Z → 25	Decryption: $((25 - 2) \times 7^{-1}) \bmod 26$	P:07 → h
C: E → 04	Decryption: $((04 - 2) \times 7^{-1}) \bmod 26$	P:04 → e
C: B → 01	Decryption: $((01 - 2) \times 7^{-1}) \bmod 26$	P:11 → l
C: B → 01	Decryption: $((01 - 2) \times 7^{-1}) \bmod 26$	P:11 → l
C: W → 22	Decryption: $((22 - 2) \times 7^{-1}) \bmod 26$	P:14 → o

Example 3.12

The additive cipher is a special case of an affine cipher in which $k_1 = 1$. The multiplicative cipher is a special case of affine cipher in which $k_2 = 0$.

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Monoalphabetic Substitution Cipher

Because additive, multiplicative, and affine ciphers have small key domains, they are very vulnerable to brute-force attack.

A better solution is to create a mapping between each plaintext character and the corresponding ciphertext character. Alice and Bob can agree on a table showing the mapping for each character.

An example key for monoalphabetic substitution cipher

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	N	O	A	T	R	B	E	C	F	U	X	D	Q	G	Y	L	K	H	V	I	J	M	P	Z	S	W

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Polyalphabetic Ciphers

In polyalphabetic substitution, each occurrence of a character may have a different substitute. The relationship between a character in the plaintext to a character in the ciphertext is one-to-many.

Autokey Cipher

$P = P_1P_2P_3 \ldots$ $C = C_1C_2C_3 \ldots$ $k = (k_1, P_1, P_2, \ldots)$

Encryption: $C_i = (P_i + k_i) \bmod 26$ Decryption: $P_i = (C_i - k_i) \bmod 26$

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Example

Assume that Alice and Bob agreed to use an autokey cipher with initial key value k_1 = 12. Now Alice wants to send Bob the message "Attack is today". Enciphering is done character by character.

Plaintext:	a	t	t	a	c	k	i	s	t	o	d	a	y
P's Values:	00	19	19	00	02	10	08	18	19	14	03	00	24
Key stream:	*12*	*00*	*19*	*19*	*00*	*02*	*10*	*08*	*18*	*19*	*14*	*03*	*00*
C's Values:	12	19	12	19	02	12	18	00	11	7	17	03	24
Ciphertext:	**M**	**T**	**M**	**T**	**C**	**M**	**S**	**A**	**L**	**H**	**R**	**D**	**Y**

$P = P_1P_2P_3 \ldots$ $C = C_1C_2C_3 \ldots$ $k = (k_1, P_1, P_2, \ldots)$

Encryption: $C_i = (P_i + k_i) \bmod 26$ Decryption: $P_i = (C_i - k_i) \bmod 26$

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Playfair Cipher

An example of a secret key in the Playfair cipher

Secret Key =

L	G	D	B	A
Q	M	H	E	C
U	R	N	I/J	F
X	V	S	O	K
Z	Y	W	T	P

Example

he → EC lx → QZ lo → BX

Plaintext: hello Ciphertext: ECQZBX

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Rules

All the letters should be written

- in capital letter,
- in pairs,
- without punctuation,
- All Js are replaced with Is.
- double letters which occur in a pair must be divided by an X or a Z.

E.g. if plain text is

- Shi Sherry loves Heath Ledger
- SH IS HE RX RY LO VE SH EA TH LE DG ER

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Cont'd

Preparing Key square (5x5)

- No repeat letter
- No Js
- KEY: SHERRY

S	H	E	R	Y
A	B	C	D	F
G	I/J	K	L	M
N	0	P	Q	T
U	V	W	X	Z

1. letters appear on the same row: **replace them with the letters to their immediate right**
2. letters appear on the same column: **replace them with the letters immediately below respectively**
3. not on the same row or column: **replace them with the letters on the same row respectively but at the other pair of corners of the rectangle defined by the original pair.**

SH IS HE RX RY LO VE SH EA TH LE DG ER

HE GH ER DR YS IQ WH HE SC OY KR AL RY

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Keyless Transposition Ciphers

- A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.

Example

A good example of a keyless cipher using the first method is the rail fence cipher. The ciphertext is created reading the pattern row by row. For example, to send the message "Meet me at the park" to Bob, Alice writes

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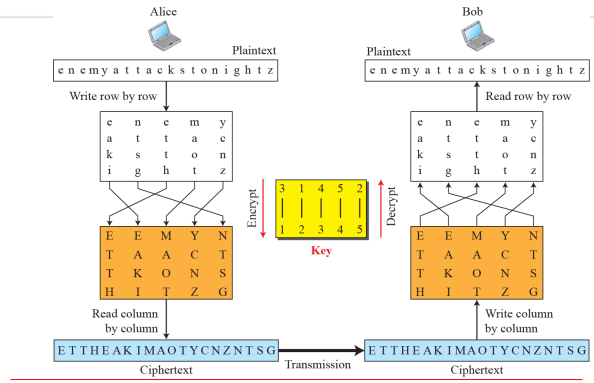
Transposition cipher



44

Cryptography vs. Cryptanalysis

- **Cryptography** is the science of designing cipher systems. Cryptographers are often known as **code-makers**.
- **Cryptanalysis** is the process of deducing information about the plaintext from the ciphertext without having knowledge of the key. Cryptanalysts are also called **code-breakers**.

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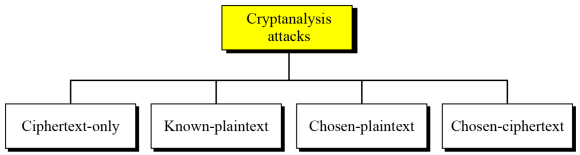
Cryptanalysis Attacks



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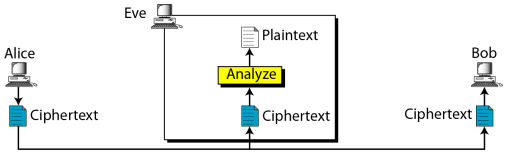
Ciphertext-Only Attack

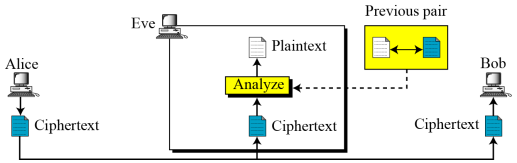
- These are most probable one because Eve needs only cipher text for this attack. Various method can be used
  - Brute-force method
    - Try all possible keys
  - Statistical Attack
  - Pattern Attack



47

Known-Plaintext Attack

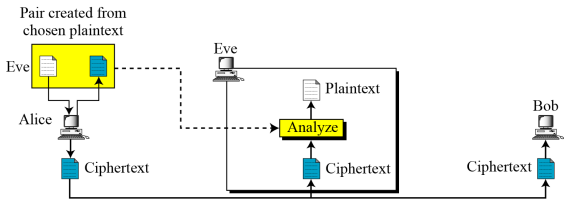
- Eve has access to both plain text and ciphertext.



48

Chosen-Plaintext Attack

- The plaintext/ciphertext pairs have been chosen by the attacker herself.



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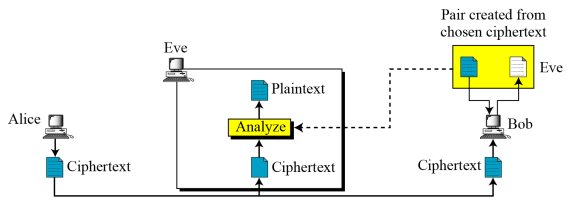
Chosen-Ciphertext Attack



50

Frequency analysis - Demonstration

- For example, let us consider the following ciphertext:

```
PCQ VMJYPD LBYK LYSO KBXBJXWXV BXV ZXJPO EYPD
KBXBJYUXJ LBJOO KCPK. CP LBO LBCMKXPV XPV IYJKL PYDBL,
QBOP KBO BXV OPVOV LBO LXRO CI SX'XJMI, KBO JCKO XPV
EYKKOV LBO DJCMPV ZOICJO BYS, KXUYPD: "DJOXL EYPD, ICJ X
LBCMKXPV XPV CPO PYDBLK Y BXNO ZOOP JOACMPLYPD LC UCM LBO
IXZROK CI GXKL XDOK XPV LBO RODOPVK CI XPAYOPL EYPDK.
SXU Y SXEO KC ZCRV XK LC AJXNO X IXNCMJ CI UCMJ SXGOKLU?"
```

- The immediate reaction of any cryptanalyst upon seeing the ciphertext is to analyse the frequency of all letters.

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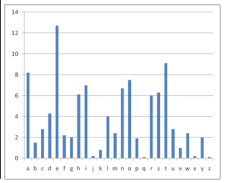
Frequency of characters in English

Letter	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*
E	12.7	H	6.1	W	2.3	K	0.08
T	9.1	R	6.0	F	2.2	J	0.02
A	8.2	D	4.3	G	2.0	Q	0.01
O	7.5	L	4.0	Y	2.0	X	0.01
I	7.0	C	2.8	P	1.9	Z	0.01
N	6.7	U	2.8	B	1.5		
S	6.3	M	2.4	V	1.0		



Frequency of diagrams and trigrams

Digram	TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF
Trigram	THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH

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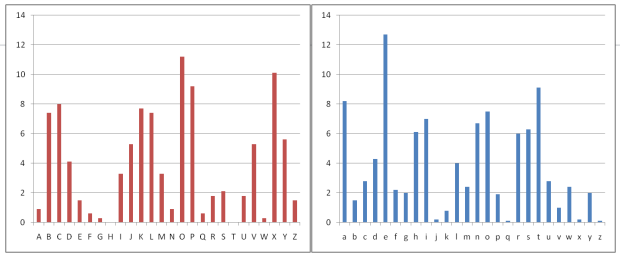
Frequency analysis - Demonstration



Frequency analysis of ciphertext Standard frequency of English letters

- An application of frequency analysis would lead to complete gibberish. We can begin by focusing only on the three most popular ciphertext letters, namely **O**, **P** and **X**.

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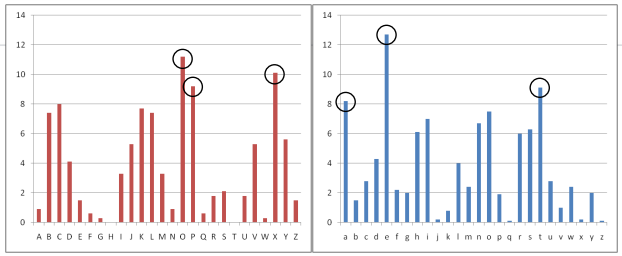
Frequency analysis - Demonstration



Frequency analysis of ciphertext Standard frequency of English letters

- Tentative assumption:

  O = **a**, **e** or **t**; **P** = **a**, **e** or **t**; **X** = **a**, **e** or **t**
- How do these letters appear next to all the other letters?

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Frequency analysis - Demonstration

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	1	9	0	3	1	1	1	0	1	4	6	0	1	2	2	8	0	4	1	0	0	3	0	1	1	2
X	0	7	0	1	1	1	1	0	2	4	6	3	0	3	1	9	0	2	4	0	3	3	2	0	0	1
P	1	0	5	6	0	0	0	0	0	1	1	2	2	0	8	0	0	0	0	0	0	11	0	9	9	0

- The table above takes the three commonest letters in the ciphertext and lists how frequently each appears before or after every letter in the cyphertext.
- Letters **O** and **X** are more 'sociable' than **P**, hence **O** and **X** represent vowels, while **P** represents a consonant.
- However, is it **O**=**a** and **X**=**e** or **O**=**e** and **X**=**a**?

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Frequency analysis - Demonstration

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	1	9	0	3	1	1	1	0	1	4	6	0	1	2	2	8	0	4	1	0	0	3	0	1	1	2
X	0	7	0	1	1	1	1	0	2	4	6	3	0	3	1	9	0	2	4	0	3	3	2	0	0	1
P	1	0	5	6	0	0	0	0	0	1	1	2	2	0	8	0	0	0	0	0	0	11	0	9	9	0

- The table above takes the three commonest letters in the ciphertext and lists how frequently each appears before or after every letter in the cyphertext.
- Letters **O** and **X** are more 'sociable' than **P**, hence **O** and **X** represent vowels, while **P** represents a consonant.
- However, is it **O**=**a** and **X**=**e** or **O**=**e** and **X**=**a**?
- Observe that combinations of **OO** appear twice, whereas **XX** does not appear at all; since **ee** appears more often than **aa** in plaintext English, it is likely that **O**=**e** and **X**=**a**.

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Frequency analysis - Demonstration

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	1	9	0	3	1	1	1	0	1	4	6	0	1	2	2	8	0	4	1	0	0	3	0	1	1	2
X	0	7	0	1	1	1	1	0	2	4	6	3	0	3	1	9	0	2	4	0	3	3	2	0	0	1
P	1	0	5	6	0	0	0	0	0	1	1	2	2	0	8	0	0	0	0	0	0	11	0	9	9	0

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
After O	1	0	0	1	0	1	0	0	1	0	4	0	1	2	2	8	0	4	1	0	0	3	0	1	1	2
Before O	0	9	0	2	1	0	1	0	0	4	2	3	0	3	1	9	0	2	4	0	3	11	2	0	0	1

- The second table shows how frequently the O, which we think represents **e**, precedes or follows all the other letters in the ciphertext.

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Frequency analysis - Demonstration

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	1	9	0	3	1	1	1	0	1	4	6	0	1	2	2	8	0	4	1	0	0	3	0	1	1	2
X	0	7	0	1	1	1	1	0	2	4	6	3	0	3	1	9	0	2	4	0	3	3	2	0	0	1
P	1	0	5	6	0	0	0	0	0	1	1	2	2	0	8	0	0	0	0	0	0	11	0	9	9	0

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
After O	1	0	0	1	0	1	0	0	1	0	4	0	1	2	2	8	0	4	1	0	0	3	0	1	1	2
Before O	0	9	0	2	1	0	1	0	0	4	2	3	0	3	1	9	0	2	4	0	3	11	2	0	0	1

- The second table shows how frequently the O, which we think represents **e**, precedes or follows all the other letters in the ciphertext.
- This trick allows us to spot the letter **h**, which frequently goes before the letter **e** (as in `the`, `then`, `they`) but rarely after **e**.
- The table suggests that **B**=**h**.

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Frequency analysis - Demonstration

- We can now replace the letters O=**e**, **X**=**a** and **B**=**h** in the ciphertext:

```
PCQ VMJYPD LBYK LYSO KBXBJXWXV BXV ZXJPO EYPD
KBXBJYUXJ LBJOO KCPK. CP LBO LBCMKXPV XPV IYJKL PYDBL,
QBOP KBO BXV OPVOV LBO LXRO CI SX'XJMI, KBO JCKO XPV
EYKKOV LBO DJCMPV ZOICJO BYS, KXUYPD: "DJOXL EYPD, ICJ X
LBCMKXPV XPV CPO PYDBLK Y BXNO ZOOP JOACMPLYPD LC UCM LBO
IXZROK CI GXKL XDOK XPV LBO RODOPVK CI XPAYOPL EYPDK.
SXU Y SXEO KC ZCRV XK LC AJXNO X IXNCMJ CI UCMJ SXGOKLU?"
```

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Frequency analysis - Demonstration

- We can now replace the letters O=**e**, **X**=**a** and **B**=**h** in the ciphertext:

```
PCQ VMJYPD LhYK LYSe KhahJaWaV haV ZaJPe EYPD
KhahJYUaJ LhJee KCPK. CP Lhe LhCMKaPV aPV IYJKL PYDhL,
QheP Khe haV ePVeV Lhe LaRe CI Sa'aJMI, Khe JCKe aPV
EYKKeV Lhe DJCMPV ZeICJe hYS, KaUYPD: "DJeaL EYPD, ICJ a
LhCMKaPV aPV CPe PYDhLK Y haNe ZeeP JeACMPLYPD LC UCM Lhe
IaZReK CI GaKL aDeK aPV Lhe ReDePVK CI aPAYePL EYPDK.
SaU Y SaEe KC ZCRV aK LC AJaNe a IaNCMJ CI UCMJ SaGeKLU?"
```

- What can you deduce about letters **Y** (which appears on its own), **L** (which appears in **Lhe**), and **P** and **V** (which appear in **aPV**)?.

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Frequency analysis - Demonstration

- For Y=i, L=t, P=n and V=d, we obtain:

```
PCQ VMJYPD LhYK LYSe KhahJaWaV haV ZaJPe EYPD
KhahJYUaJ LhJee KCPK. CP Lhe LhCMKaPV aPV IYJKL PYDhL,
QheP Khe haV ePVeV Lhe LaRe CI Sa'aJMI, Khe JCKe aPV
EYKKeV Lhe DJCMPV ZeICJe hYS, KaUYPD: "DJeaL EYPD, ICJ a
LhCMKaPV aPV CPe PYDhLK Y haNe ZeeP JeACMPLYPD LC UCM Lhe
IaZReK CI GaKL aDeK aPV Lhe ReDePVK CI aPAYePL EYPDK.
SaU Y SaEe KC ZCRV aK LC AJaNe a IaNCMJ CI UCMJ SaGeKLU?"a
```

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Frequency analysis - Demonstration

- For Y=i, L=t, P=n and V=d, we obtain:

```
PCQ VMJiPD LhiK LiSe KhahJaWaV haV ZaJPe EiPD
KhahJiUaJ LhJee KCPK. CP Lhe LhCMKaPV aPV IiJKL PiDhL,
QheP Khe haV ePVeV Lhe LaRe CI Sa'aJMI, Khe JCKe aPV
EiKKeV Lhe DJCMPV ZeICJe hiS, KaUiPD: "DJeaL EiPD, ICJ a
LhCMKaPV aPV CPe PiDhLK i haNe ZeeP JeACMPLiPD LC UCM Lhe
IaZReK CI GaKL aDeK aPV Lhe ReDePVK CI aPAiePL EiPDK.
SaU i SaEe KC ZCRV aK LC AJaNe a IaNCMJ CI UCMJ SaGeKLU?"
```

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Frequency analysis - Demonstration

- For Y=i, L=t, P=n and V=d, we obtain:

```
PCQ VMJiPD thiK tiSe KhahJaWaV haV ZaJPe EiPD
KhahJiUaJ thJee KCPK. CP the thCMKaPV aPV IiJKt PiDht,
QheP Khe haV ePVeV the taRe CI Sa'aJMI, Khe JCKe aPV
EiKKeV the DJCMPV ZeICJe hiS, KaUiPD: "DJeat EiPD, ICJ a
thCMKaPV aPV CPe PiDhtK i haNe ZeeP JeACMPtiPD tC UCM the
IaZReK CI GaKt aDeK aPV the ReDePVK CI aPAiePt EiPDK.
SaU i SaEe KC ZCRV aK tC AJaNe a IaNCMJ CI UCMJ SaGeKtU?"
```

63

Frequency analysis - Demonstration

- For Y=i, L=t, P=n and V=d, we obtain:

```
nCQ VMJinD thiK tiSe KhahJaWaV haV ZaJne EinD
KhahJiUaJ thJee KCnK. Cn the thCMKanV anV IiJKt niDht,
Qhen Khe haV enVeV the taRe CI Sa'aJMI, Khe JCKe anV
EiKKeV the DJCMnV ZeICJe hiS, KaUinD: "DJeat EinD, ICJ a
thCMKanV anV Cne niDhtK i haNe Zeen JeACMntinD tC UCM the
IaZReK CI GaKt aDeK anV the ReDenVK CI anAient EinDK.
SaU i SaEe KC ZCRV aK tC AJaNe a IaNCMJ CI UCMJ SaGeKtU?"
```

64

Frequency analysis - Demonstration

- For Y=i, L=t, P=n and V=d, we obtain:

```
nCQ dMJinD thiK tiSe KhahJaWad had ZaJne EinD
KhahJiUaJ thJee KCnK. Cn the thCMKand and IiJKt niDht,
Qhen Khe had ended the taRe CI Sa'aJMI, Khe JCKe and
EiKKed the DJCMnd ZeICJe hiS, KaUinD: "DJeat EinD, ICJ a
thCMKand and Cne niDhtK i haNe Zeen JeACMntinD tC UCM the
IaZReK CI GaKt aDeK and the ReDendK CI anAient EinDK.
SaU i SaEe KC ZCRd aK tC AJaNe a IaNCMJ CI UCMJ SaGeKtU?"
```

65

Frequency analysis - Demonstration

- For Y=i, L=t, P=n and V=d, we obtain:

```
nCQ dMJinD thiK tiSe KhahJaWad had ZaJne EinD
KhahJiUaJ thJee KCnK. Cn the thCMKand and IiJKt niDht,
Qhen Khe had ended the taRe CI Sa'aJMI, Khe JCKe and
EiKKed the DJCMnd ZeICJe hiS, KaUinD: "DJeat EinD, ICJ a
thCMKand and Cne niDhtK i haNe Zeen JeACMntinD tC UCM the
IaZReK CI GaKt aDeK and the ReDendK CI anAient EinDK.
SaU i SaEe KC ZCRd aK tC AJaNe a IaNCMJ CI UCMJ SaGeKtU?"
```

- Once a few letters have been established, cryptanalysis progresses very rapidly!

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Frequency analysis - Demonstration

- The unscrambled ciphertext is:

```
now during this time shahrazad had borne king
shahriyar three sons. on the thousand and first night,
when she had ended the tale of ma'aruf, she rose and
kissed the ground before him, saying: "great king, for a
thousand and one nights i have been recounting to you the
fables of past ages and the legends of ancient kings.
may i make so bold as to crave a favour of your majesty?"
```

Epilogue, *Tales from the Thousand and One Nights*.

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Thank you

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